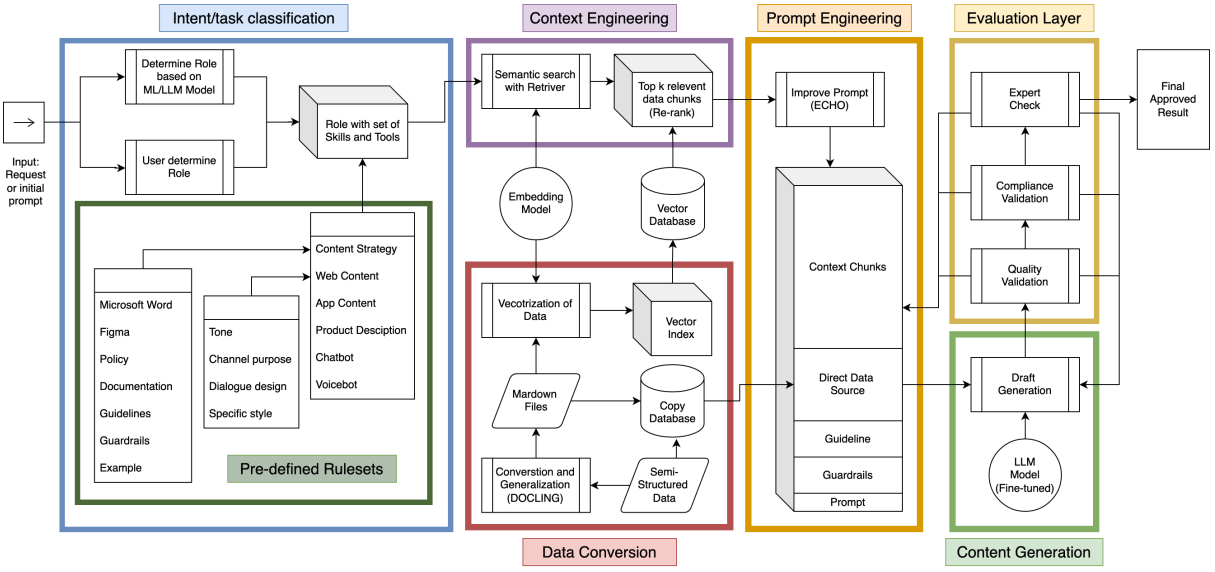
Project Overview

Status	Notes
Assign	Adam Dong
AI keywords	
Topic	General

Project Overview — ABN AMRO AISO Editorial Co-pilot (working title: AURORA)

> **For:** team members, ABN AMRO stakeholders, and future contributors.
> **Purpose:** single canonical reference describing the project's goal, architecture, current stage, and roadmap.



1. TL;DR

We are building an **AI-powered editorial co-pilot** for the ABN AMRO Insights publication channel. The system helps domain experts — who are not

professional content writers — produce articles that are consistent with ABN AMRO's tone of voice, writing guidelines, and approved examples.

The deliverable to ABN AMRO is **two-fold**:

1. **A working POC** that demonstrates retrieval-grounded drafting and review.
2. **A business analysis** that compares two viable retrieval architectures across cost, complexity, maintenance, security, retrieval quality, scalability, explainability, vendor lock-in, and data freshness — so ABN AMRO can choose the right path before they invest in production.

We are deliberately implementing **both** retrieval architectures (vector-based and vectorless / PageIndex tree-based) so the comparison is grounded in measurement, not speculation.

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2. Project Goal & Scope

2.1 Business problem

The Insights page on the ABN AMRO website hosts articles written by experts in their domain (banking, sector analysis, technology). These experts are subject-matter authorities, but **not content writers**. As a result:

- Drafts vary in tone and structure.
- Manual review is a bottleneck for the content strategy team.
- Aging articles need periodic refresh, but tracking which ones is manual.
- The system is also migrating to a new CRM, so an automated quality gate before ingestion is timely.

2.2 What the co-pilot does (the five tasks)

ID	Task	What it means
T1	Help experts create content that aligns with the criteria	Drafting assistance grounded in writing guide + approved examples
T2	Check articles before they enter the new CRM database	Automated quality + compliance validation
T3	Compose checks into an automated pipeline / agent skill	Plug-and-play orchestration, reusable across channels
T4	Renew articles older than one	Detect aging content and propose

ID	Task	What it means
	year	refreshes
T5	Provide good business analysis for ABN AMRO	**First-class deliverable** — document tradeoffs, decisions, costs

T5 is not a side note. It elevates the project from “a POC” to “a POC + a buy/build/operate decision”.

2.3 Goal shift from the original framing

The original project name *AURORA — Autonomous Unified Reasoning & Output Review Agent* leaned heavily on **review**. The current scope is broader: drafting + review + refinement + maintenance, with a human always in the loop. Treat the co-pilot as a partner to the editor, not a gatekeeper.

> **Naming note.** *AURORA* is a working title only. Given the broadened scope, candidate replacements include **SCRIBE** (Style-Consistent Retrieval-Informed Banking Editorial), **MUSE** (Multi-source Unified Support for Editorial), or a re-backronymed *AURORA — Aligned Unified Retrieval & Refinement Of Approved Articles*. Naming is not load-bearing — finalize when the architecture lands.

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3. Architecture — Two Parallel Tracks

The architecture diagram in `docs/ABNAMRO.drawio.svg` shows the **target conceptual pipeline**. Within that pipeline, the **retrieval layer** is the most consequential decision, and we are evaluating two implementations in parallel.

3.1 Shared backbone (independent of retrieval choice)

```
┌──────────┐    ┌──────────────────┐    ┌──────────────────┐    ┌──────────┐
│  Input   │ → │   Role Selection   │ → │   Retrieval Layer  │ → │  Prompt  │ → Draft / Review
│ (prompt) │    │ (LLM picks role)   │    │   (Track A or B)   │    │ Assembly │
└──────────┘    └──────────────────┘    └──────────────────┘    └──────────┘
                         |                                            |
                         ▼                                            ▼
                Skills + Tools/Content                      + Guidelines + Guardrails
                (Tone, Channel purpose,                     + ECHO prompt-improvement loop
                 Dialogue design, Style)
```

Roles (the system chooses one based on the request): Content Strategy, Web Content, App Content, Product Description, Chatbot, Voicebot.
Skills: Tone, Channel purpose, Dialogue design, Specific style.

Tools/Content sources: MS Word, Figma, Policy, Documentation, Guidelines, Guardrails, Examples.
ECHO: prompt-improvement step that combines original prompt + retrieved context + guidelines + guardrails into the final LLM call. (Note: ECHO is a theoretical framework, which is literally prompt engineering in ABN AMRO context)

3.2 Track A — Vector-based RAG (target architecture in the diagram)

- Documents are chunked and embedded.
- A vector index (FAISS / pgvector / managed service) supports semantic search.
- Retriever returns top-k chunks; a re-ranker refines order.
- **Owners:** Ilgara (chunking + index design), Gaoxiang (embedding model + similarity metric, co-optimised with chunk size).

3.3 Track B — Vectorless / PageIndex tree retrieval (current POC)

- Documents are converted to a hierarchical tree (PageIndex) with node-level summaries.
- Document selection is **description-based** against a small registry (`corpus_manifest.json`).
- Section retrieval scores tree nodes by overlap with the prompt (title, summary, text).
- No embedding spend, no vector DB, no re-indexing on every doc change.
- **Owner:** Adam (PageIndex investigation). Live POC in `pageindex_api/` and `test_pageindex/`.

3.4 Hybrid possibility

A natural endpoint is a **hybrid**: PageIndex for normative documents (Writing Guide — small, hierarchical, authoritative) and vectors for the growing corpus of approved example articles (large-N, semantic similarity matters). The comparison matrix will inform whether the hybrid earns its complexity.

3.5 Architecture comparison matrix

This is the core of the T5 deliverable. Cells will be filled by measurement and analysis at the end of Stage 1.

Dimension	Track A — Vector RAG	Track B — PageIndex tree	Notes / weight for ABN AMRO
Cost	TBD — embedding spend + vector DB hosting	TBD — PageIndex API calls only	Recurring infra is a real line item
Complexity	TBD — more components	TBD — fewer moving parts	Affects ABN AMRO's ability to operate it in-house
Maintenance	TBD — re-embed on doc updates	TBD — re-tree on doc updates	Cadence depends on how often Insights publishes
Security & data residency ⚑	TBD — depends on vector DB choice (EU region, on-prem option)	TBD — PageIndex SaaS data path	**High weight**: banking + EU sovereignty (also a theme in the corpus itself)
Retrieval quality	TBD — recall@k, faithfulness	TBD — section-level precision	Measured against held-out approved articles
Scalability	Scales well to 10k+ docs	Selection model strains as corpus grows	Insights corpus is small today, growing tomorrow
Explainability / auditability ⚑	Harder — embeddings are opaque	Easier — tree path is human-readable	**High weight**: regulated banking context
Vendor lock-in ⚑	Depends on chosen DB / embedding model	Currently tied to PageIndex SaaS API	**High weight**: bank procurement & exit risk
Data freshness	New articles need re-embedding	New articles need re-tree	Both roughly equal

⚑ = ABN-AMRO-specific high-weight dimension.

4. Data Sources

Source	Type	Use
Insights page articles (approved)	Markdown — currently 5 nl + their en translations	Approved examples, authority rank ~60
Writing Guide 2026 V1.1	PDF	Normative authority, authority rank 100
Internal versions / rejected drafts	*To be provided*	Anti-pattern examples for Track B's "rejected example" track
Checklist + KPI list	*To be provided*	Drives evaluation metrics (T2, T5)

The corpus is intentionally small in Stage 1 so we can iterate on retrieval quality without being slowed down by infra concerns. See `test_pageindex/corpus_manifest.json` for the live registry.

5. Current State (Stage 1, deadline 2026-05-11)

Area	Status	Where it lives
PageIndex POC for vectorless retrieval	**Live** — `select_and_retrieve.py` runs end-to-end	`test_pageindex/`
PageIndex API client wrapper	In progress — tree fetch + document upload	`pageindex_api/main.py`
Article corpus (en + nl)	5 approved + Writing Guide indexed	`data/article/{en,nl}/`
Vector RAG track	Research phase — chunking + embedding investigations	(Ilgara / Gaoxiang notes)
Evaluation methodology	Defined: Reviewer Acceptance Rate + compliance score	(El Yassae notes)
Task definition / channel dataflows	Research phase	(Yuvraj notes)

The POC's frontend is intentionally minimal — a CLI / bash entry point. No web UI is in scope until the architecture is decided.

6. Team & Roles

Member	Focus area
Adam Dong	Rule profile for insight content; initial datasource gathering; PageIndex / vectorless RAG investigation

Member	Focus area
El Yassae	Evaluation layer (metrics, KPIs); context engineering — chunk composition, re-rank logic, bridge between Data Conversion and Prompt Engineering
Yuvraj	Task definition mechanism; channel-specific dataflows; service best practices; plug-and-play design
Ilgara Yusifzada	Data source investigation and conversion; chunking strategy comparison; vector DB index design
Gaoxiang	Embedding model + similarity metric; co-optimisation of chunk count and chunk size

7. Roadmap (post Stage 1)

Stage 2 — convergence

- Lock the retrieval architecture (A, B, or hybrid) based on the measurement matrix in §3.5.
- Promote document selection from description-based to a real metadata layer.
- Add rejected drafts as anti-pattern documents (depends on data drop).
- Wire the checklist + KPI list into automated quality + compliance validation (T2).

Stage 3 — productisation

- Wrap the flow in a thin FastAPI or Streamlit interface for content-strategy users.
- Per-run audit trail: prompt → selected docs → selected sections → final context pack → LLM output → reviewer decision.
- Skill-based packaging (T3) so the same engine can serve other channels (App Content, Chatbot, Voicebot).
- Article aging detector for the renewal workflow (T4).

Open questions / decisions pending

- Vector DB choice (managed SaaS vs self-hosted, EU residency).
- Embedding model (proprietary OpenAI vs open-source, multilingual nl/en).

- Where the LLM runs (Azure OpenAI in EU, on-prem, etc.) — security-driven.
- How rejected drafts are surfaced without being treated as authoritative.
- Whether T4 (aging) reuses the same retrieval pipeline or is a separate scheduled job.